

THE ARCTIC OSCILLATION INFLUENCE ON INDIAN SUMMER MONSOON RAINFALL

Project Report in Partial Fulfilment for the Award of the Degree of

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DECLARATION

I hereby declare that the Project report entitled “**The Arctic Oscillation (AO) Influence on Indian Summer Monsoon Rainfall**” submitted for the partial fulfilment of M. Tech degree in Atmospheric Sciences, Department of Atmospheric and Space Sciences, Savitribai Phule Pune University is carried out by me at Indian Institute of Tropical Meteorology Pune and the Department of Atmospheric and Space Sciences, Savitribai Phule Pune University. Under the guidance of **Dr. Jasti S Chowdary** and **Dr. Aditi Deshpande**. This work or any part of it has not been submitted to any other institution/university for the award of any degree/diploma. I further declare that the material obtained from other sources has been duly acknowledged in the project report.

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CERTIFICATE

This is to certify that the Project report entitled “**The Arctic Oscillation (AO) Influence on Indian Summer Monsoon Rainfall**” submitted by **Adarsh Dattarao Raut** for the fulfilment of the M. Tech degree in Atmospheric Sciences at the Department of Atmospheric and Space Sciences, Savitribai Phule Pune University has been carried out by him under my supervision at the Indian Institute of Tropical Meteorology, Pune and Department of Atmospheric and Space Sciences, SPPU, Pune.

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ABSTRACT

The Indian Summer Monsoon Rainfall (ISMR) significantly impacts the economies and livelihoods of millions in India. This study investigates the influence of the Arctic Oscillation (AO) on the ISMR, acknowledging its crucial role alongside other climatic drivers such as El Niño Southern Oscillation (ENSO). Through an extensive analysis of atmospheric and oceanic data spanning from 1950 to 2023, utilizing the European Centre for Medium-Range Weather Forecasts (ECMWF) reanalysis ERA5 and Global Precipitation Climatology Centre (GPCC) datasets, alongside Arctic Oscillation Index (AOI) calculations, the study explores correlations between AO phases and ISMR variability. Findings suggest a notable connection between AO phases and ISMR, particularly evident in positive AO phases impacting precipitation patterns across mid-latitudes and the Central Indian region. Additionally, the study reveals consistent correlations between AO phases and atmospheric parameters such as geopotential height, sea level pressure, and wind patterns, further emphasizing the AO's influence on the ISMR. Overall, this research contributes to a deeper understanding of the complex interactions between large-scale climatic oscillations and regional monsoon dynamics, crucial for enhancing predictive capabilities and mitigating the socioeconomic impacts of monsoon variability.

Chapter 1

Introduction

1.1 Indian Summer Monsoon Rainfall

In the Indian subcontinent, low level winds appear to blow northeasterly during the boreal winter season and south-westerly during the summer season from June to September (Simpson, 1921; Ramage, 1971; Rao, 1976; Webster, 1987; Das et al., 1968) as shown in figure 1.1. The summer monsoon accounts for more than 75% of the annual rainfall over India (e.g., Parthasarathy et al., 1994) however, the amount of seasonal mean rainfall shows remarkable variability from sub-seasonal to decadal time scale. Studies showed that even a few minor changes in its spatio-temporal variability of this Indian Summer Monsoon (ISM) rainfall can bring substantial hardships (e.g., Sinha et al., 2011) shaping the socioeconomic condition and livelihood of billions of peoples of India. In an agricultural country like India, the extreme departure from normal seasonal rainfall (droughts and floods) seriously affects agricultural output and thus the economy of the country (e.g., Mooley and Parthasarathy 1984; Webster et al.1998; Kripalani et al. 2003; Chaudhari et al. 2010). Therefore, reliable prediction of ISM rainfall and its year-to-year variation has immense importance (e.g., Pokhrel et al., 2012). ISM rainfall is a large land–ocean–atmosphere coupled system (Loschnigg and Webster 2000; *Narapusetty et al., 2016*).. It occurs due to the transport of substantial amounts of moisture during boreal summer across the Indian Ocean toward the Indian subcontinent (e.g., Webster et al. 1998). It reaches as far north as the southern Himalayas and as far northwest as Pakistan (Gayatri et al., 2017).

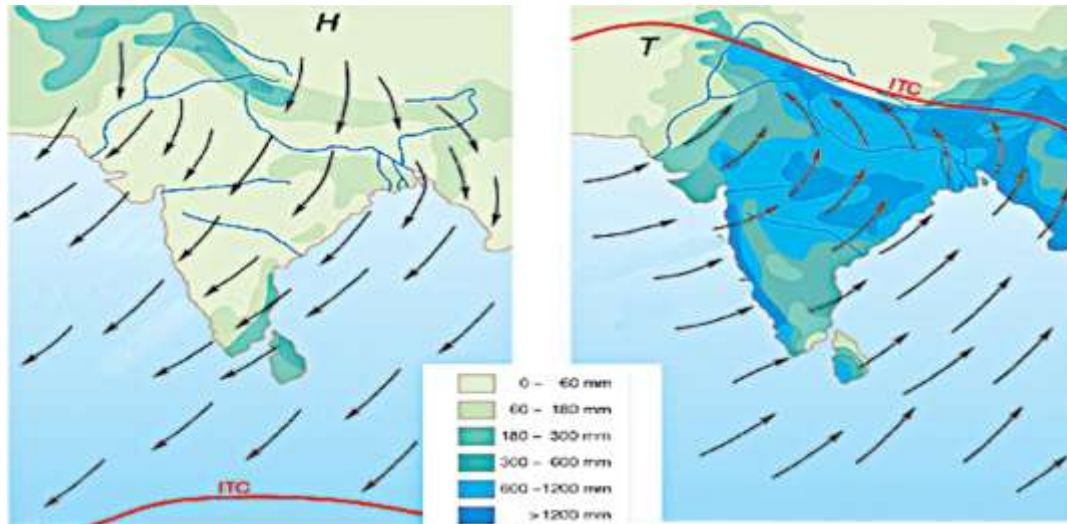


Figure 1.1: Indian monsoon with seasonal reversal of wind system over the Indian subcontinent and surrounding oceans for summer (southwest) and winter (northeast) monsoon.(source: <https://insider-week.com>)

1.2 Driving factors of ISM rainfall

The monsoon variability depends upon several factors such as internal dynamics, influences of land and ocean variability, and teleconnections to climate variability in other regions (e.g., Goswami et al., 1999; Webster et al., 1998; Krishna Kumar et al., 1999). Further, a myriad of large-scale atmospheric, oceanic, and coupled climate phenomena influence the spatio-temporal complexity of ISM rainfall variability, including monsoon depressions, Madden–Julian Oscillation, El Niño–Southern Oscillation (ENSO), Indian Ocean Dipole (IOD), Indian Ocean capacitor, Eurasian snow cover changes, the Pacific Decadal Oscillation, the Atlantic Multidecadal Oscillation, and anthropogenic climate change (e.g., Chowdary et al., 2021). A deep physical understanding of these modes and their interactions with the ISM is the foundation for improved predictions and projections from interannual to decadal timescales (Vibhute et al., 2020, 2023). Major sources of predictability of ISM rainfall on interannual timescales comes from the slowly varying large-scale boundary conditions (Charney and Shukla, 1981; Shukla, 1998.). The equatorial Pacific sea surface temperature (SST) anomalies associated with the ENSO are one such source (Goswami et al., 2023). Previous studies based on linear diagnostic analysis revealed that only about 50% of droughts in India are associated with ENSO (Kripalani and Kulkarni 1996). Various researchers have shown that the interannual variability (IAV) is influenced by several other elements of the climate system majorly, snow cover variations over Eurasia (Blanford 1884; Hahn and Shukla 1976;

Bamzai and Shukla 1999; Kripalani and Kulkarni 1999), North Atlantic Ocean circulation (Goswami et al. 2006; Yadav 2009a, b), IOD (Saji et al. 1999; Ashok et al. 2001), Equatorial Indian Ocean Oscillation (EQUINOO; Gadgil et al. 2007), Atlantic zonal mode (Sabeerali et al., 2019), Subtropical westerly jet (Chowdary et al., 2022; Vibhute et al., 2023), Arctic Oscillation (AO) (change et al., 2001; Kakade and Kulkarni, 2017), etc.,. The extratropical systems influence the ISM rainfall via Rossby wave breaking along the upper-level westerly jet also known as the Asian jet (e.g., Samanta et al., 2016). Previous studies suggested that the meridional displacement of the Asian jet also alters the interannual rainfall over India and East Asian region during summer by interacting with tropical circulation (e.g., Lau et al., 1988, Lau et al., 2000; Chowdary et al., 2022). Previous studies also reported that the western north Pacific (WNP) anticyclone exhibits a significant influence on ISM rainfall (e.g., Kosaka et al., 2013; Chowdary et al., 2017, 2019; Zhou et al., 2019; Darshana et al., 2020). The WNP anticyclone is a part of the positive Pacific-Japan (PJ) pattern (Nitta, 1987) that plays an important role in East Asian as well as Indian summer monsoon variability (e.g., Huang and Yf, 1989; Zhang et al., 1996; Wang et al., 2000; Srinivas et al. 2018). This WNP circulation variability is significantly related to ENSO (e.g., Wang et al., 2003; Xie et al., 2009; Li et al., 2017).

Apart from tropics and subtropical climate modes, extratropical snow cover, Arctic Amplification, southern angular mode and northern angular mode, North Atlantic Oscillation (NAO) etc. are known to impact the climate over India in various seasons (e.g., Goswami et al., 2006; Roy, 2011). The influence of Arctic Oscillation on Indian winter monsoon is studied by Midhuna and Dimri (2019). Further, association between Arctic Circulation and ISM Rainfall is studied by Kakade and Kulkarni (2017). Recently, Sandhya and Sridharan (2022) pointed out that February AO and ISM rainfall during 1957–1977 and 1978–1998 correlated positively, and the correlation changed to negative during 1999–2018. However, simultaneous relationship or influence of AO in boreal summer on ISM rainfall is not well studied. Further, mechanisms by which AO modulates the ISM rainfall is also explored in this report. Thus, the focus of the study is to understand the impact of Arctic Oscillation on ISM rainfall. It is important to primarily know about the AO as introduced in the next sub-section.

1.3 Arctic Oscillation (AO) and its global influence.

The Arctic Oscillation (AO) is a see-saw pattern in an atmospheric circulation between the Polar region and mid-latitudes of the Northern Hemisphere, also referred to as the Northern Hemisphere annular mode (Dahlman 2009; Deser 2000; Feldstein and Franzke, 2006: Fig. 1.2). It is known that there is low pressure centered over the North Pole. Previous studies reported that this tropospheric vortex tends to be stronger in winter than it is in summer but persists throughout the year (Lawrence et al., 2020). The most dominant phase of this oscillation can alter the north-to-south storm-steering and mid-latitude jet stream. Therefore, this AO has a strong influence on weather and climate in major population centres in North America, Europe, and Asia (e.g., Shengping et al., 2015).

AO can be divided into the positive and negative phases.

The positive phase (Figure 1.2a) occurs when the polar vortex has a strong low-pressure system and the mid-latitudes have a strong high-pressure system (NOAA Climate.gov). This led to a stronger and straighter northward shifted polar jet along with northward shifted westerly jet over northern Tibetan Plateau (He et al., 2017; Mao et al., 2011a; Mao et al., 2011b), resulting in arctic air locking in the polar region. It is evident that this positive AO prevents penetration of cold Arctic air from polar to mid-latitudes and causes warmer winters over much of the mid-latitudes and East Asia (e.g., Thompson and Wallace, 2000; Sandhya and Shridharan, 2022).

On the other hand, in the negative phase (Figure 1.2b) of the arctic oscillation high pressure is confined at the polar region penetrating into the mid-latitudes, leading to weakening of the westerly jet, allowing cold air to enter the sub-tropical zone (Sandhya and Shridharan, 2022). The polar jet is also feeble, which meanders and frequently stretches meridional far to the north and south with cold air that can periodically escape from the Arctic (Maeda et al., 2000; Tachibana et al., 2010; Nakamura and Fukamachi 2004; Tachibana et al., 2019; Sandhya and Shridharan, 2022) and influences the mid latitude weather.

Many studies also conclude that a positive winter AO could lead to a weaker-than-normal East Asian Winter Monsoon (EAWM) and warm conditions over East Asia through its impacts on the Siberian high (e.g., Gong et al., 2001, Wu and Wang, 2002a), Rossby wave (Park et al., 2011), and the polar and subtropical jets (Li et al., 2014). Arctic Oscillation influences on sea ice and marine productivity of northwest Greenland over the last 9,000 years (Audrey et al., 2020) and storm variability (Feser et al., 2015). The impact of winter AO on Eurasian Surface Air Temperature (SAT) and cold surges shows decadal change (e.g.,

Woo et al., 2012). Studies also suggest that during the positive phase of the AO, Positive SAT anomalies elongated over Europe to East Asia, while strong cold anomalies appeared over Greenland (Woo et al., 2012). However, the amplitude of positive SAT anomalies related to the positive phase of AO was larger in the 1990s and 2000s than 1980s (Woo et al., 2012). Much literature is found on AO's influence over the globe during winter time but summertime AO variability and its global influence cannot be ignored. Zhou et al., (2024) analyzed that changes in the summertime AO pattern can alter the wave activity flux over the Eurasian continent. The zonal wave train associated with AO can serve as a bridge to transport the distinct impacts of the summer AO eastward to East Asia. Presence of significant correlation exists between summertime AO on ISM rainfall, only few studies are found which depict the influence of summertime AO on ISM rainfall which are discussed in the next chapter.

The Arctic Oscillation Index (AOI) is calculated as the first leading mode (which captures the maximum amount of variance) from the Empirical Orthogonal Function (EOF) analysis of monthly mean geopotential height anomalies at 1000-hPa, poleward of 20°N in the Northern hemisphere (Zhang et al., 2021). It can also be obtained from the Climate Prediction Centre (CPC) website (cpc.ncep.gov/products/precip.CWlink./daily_ao_index) from the years 1950 - 2023 (Thompson et al., 1998).

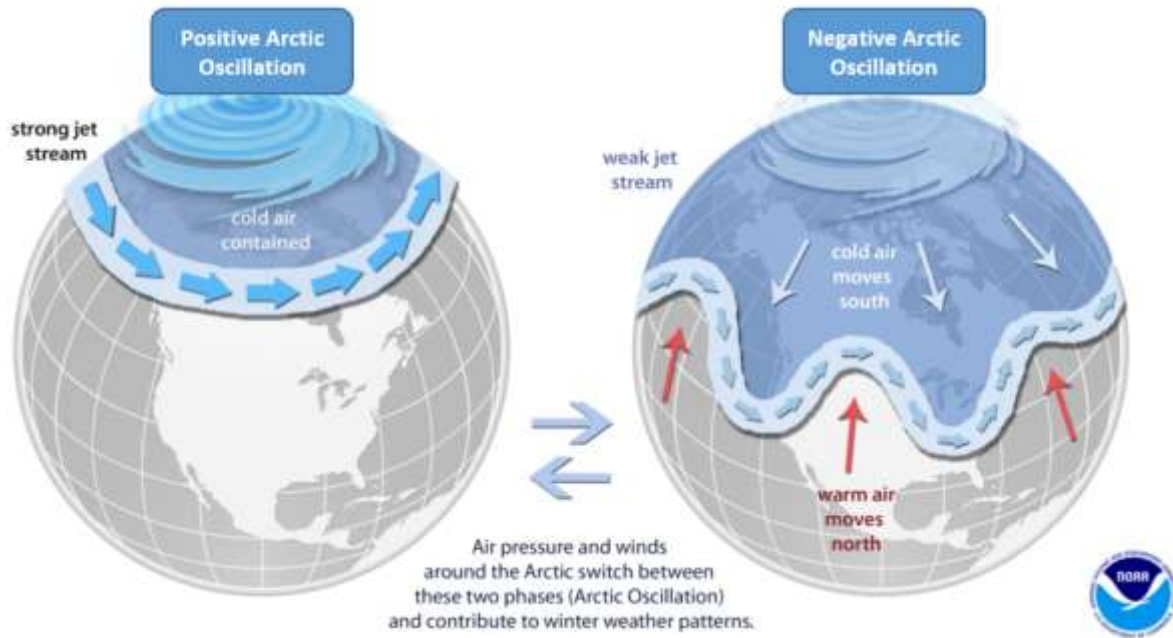


Figure 1.2: A schematic diagram showing the characteristics of the polar jet during the positive and negative phases of the Arctic Oscillation. (source :

<https://www.worldclimateservice.com/2021/09/01/arctic-oscillation/>)

As shown in figure 1.2 Lower-than-average air pressure over the Arctic and higher-than-average pressure across the northern Pacific and Atlantic Oceans are characteristics of the positive phase of the AO. Storms can be pushed north of their typical trajectories because the jet stream is farther north than usual in these conditions. Consequently, during the positive phase of the AO, there are typically fewer cold air outbreaks than usual across the mid-latitudes of North America, Europe, Siberia, and East Asia (Zhang et al.,2022).

On the other hand, the Arctic region experiences higher-than-average air pressure during AO's negative phase, while the northern Pacific and Atlantic Oceans have lower-than-average pressure. In these conditions, the jet stream moves toward the equator, causing the river of air encircling the globe to be south of its typical position. As a result, during winters when the AO is negative, mid-latitude regions are more prone to witness outbreaks of cold, polar air. For instance, in New England, the negative phase of AO is associated with increased frequency of coastal storms called "Nor'easters" (Shengping et al.,2017). The Antarctic Oscillation (AAO), a comparable pattern of air pressure and jet stream anomalies in the Southern Hemisphere, is comparable to AO phases. These patterns exhibit a distinctive ring-shaped or "annular" pattern when viewed from above either pole; for this reason, AO and AAO are also known as the Northern Annular Mode (NAM) and Southern Annular Mode (SAM), respectively (Laila et al., 2005).

Chapter 2

Literature Review

2.1 Emphasis on studies analysing the AOI influence on ISM rainfall

Many studies reported the positive correlation between AOI and ISM rainfall (for e.g., Kakade and Kulkarni, 2017; Goswami et al., 2006; Roy, 2011). Kakade and Kulkarni, (2017) analysed that AO and ISM rainfall direct concurrent relationship with each other in the summer monsoon season but mechanisms that connect AO and ISM rainfall are not well examined. The positive correlation between AO and ISMR can be interpreted as: when AO is positive then the southward penetration of Arctic cold air-mass is restricted, which may result in reducing Eurasian snow-cover and more rainfall over India (Kulkarni and Kripalani 1998; Jarvis and Patrick 1973; Bamzai and Shukla 1999). Kakade and Kulkarni, (2017) found that the circulation parameters over the Arctic can be good predictors for ISM rainfall. Sandhya and Sridharan (2022) analysed that February AOI shows a significant correlation with ISM rainfall, but this relationship was positive before 1998 and became negative after that. The mechanism behind it is that before 1998 the positive phase of AOI locks the Arctic air into the high latitudes and it can decrease the Eurasian snow cover. Hence, there is an increase in the surface temperature over the Eurasian region leading to an increase in the ISM rainfall but this relationship reverses after 1998. The North Annular mode (NAM) produces persistent weakening (strengthening) of the meridional gradient of tropospheric temperature (TT) by setting up negative (positive) TT anomaly over Eurasia during northern late summer/autumn resulting in early (late) withdrawal of the southwest monsoon and persistent decrease (increase) of seasonal monsoon rainfall (Goswami et al., 2006). These are few studies that suggest AO has association with ISM rainfall through modulation of NAO and Eurasian temperatures; however, this relationship is still a subject of debate and hence here we have examined new pathways in the present study. It is shown that the well-known positive relation between AOI and ISM seems to be shifting to negative along with the increasing trend of the AOI during the recent decades (Sandhya and Shridharan,.2022).

2.2 Research gap

The limited literature is found that analyses the association of summertime AO with ISM rainfall, their pathways and related mechanisms. Further AO association with ISM rainfall through mid-latitude tropical circulation and a subtropical westerly jet (also called Asian jet) remains to be explored.

2.3 Objective

The above mentioned research gap motivated us to analyse the various new pathways and related mechanisms of AO's influence on ISM rainfall through mid-latitude tropical circulation and a subtropical westerly jet.

Here it is aimed to analyse the pathways and related mechanisms of AO's influence on ISMR through mid-latitude tropical circulation.

Chapter 3

Datasets and Methodology

3.1 Datasets used

For Present study i have used several datasets such as atmospheric data ,Oceanic data and few indexes (i.e arctic Oscillation index and Nino 3.4 index)

3.1.1 Atmospheric Data

The fifth generation of the European Centre for Medium-Range Weather Forecasts (ECMWF) reanalysis ERA5 (Hersbach et al.,2020) monthly averaged data on a single level (e.g., Geopotential height, sea level pressure(SLP), zonal and meridional winds, land temperature) with a horizontal resolution of $0.25^{\circ} \times 0.25^{\circ}$ from 1950 to 2023 are used for the detailed analysis.

Monthly rainfall data from Global Precipitation Climatology Centre(GPCC) monthly precipitation version 2.3 (Adler et al., 2018) on a $0.25^{\circ} \times 0.25^{\circ}$ resolution and IMD precipitation data (Pai et al., 2014; Rajeevan et al.,2006) with resolution of $0.25^{\circ} \times 0.25^{\circ}$ are also used. The above dataset seems to be comprehensive and covers all atmospheric variables used for detail analysis which are useful for atmospheric research. These data are widely used for climate studies or climate research. This data is a very important source for climate study, and weather forecasting as well as for understanding long-term trends.

3.1.2 Oceanic Data

Sea Surface Temperature (SST) is obtained from the latest version of Extended Reconstructed Sea Surface Temperature (ERSST), version 5 with a spatial resolution of $2^{\circ} \times 2^{\circ}$ (Huang et al., 2017). And Niño 3.4 from the National Centre for Atmospheric Research (NCAR). SST represents the surface temperature of the ocean. It is the reconstructed data, which means it is the combined data set of observed data and statistically created to fill the gap between datasets leading to create continuous datasets. Niño 3.4 data is specified for the

particular region of Ocean surface temperature. As Niño 3.4 NCAR (National Centre for Atmospheric Research) has defined several more Niño regions based on their criteria of Niño region.

3.2 Methodology

3.2.1 Linear correlation and regression analysis

Correlation coefficients are indicators of the strength of the linear relationship between two different variables, x and y. The Pearson's correlation coefficient (r) between two variables X and Y is given by :

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Where,

n is sample size

x_i, y_i are the individual sample points indexed with i.

$\bar{x}$ and $\bar{y}$ are the climatology values.

The possible range of values for the correlation coefficient is -1.0 to 1.0. In other words, the values cannot exceed 1.0 or be less than -1.0. A correlation of -1.0 indicates a perfect negative correlation and a correlation of 1.0 indicates a perfect positive correlation. If the correlation coefficient is greater than zero, it is a positive relationship. Conversely, if the value is less than zero, it is a negative relationship. A value of zero indicates that there is no relationship between the two variables.

Linear regression is a statistical model which estimates the linear relationship between a scalar response and one or more explanatory variables (also known as dependent and independent variables). The simplest form of linear regression involves only one independent variable and one dependent variable. The equation for simple linear regression is:

$$y = \beta_0 + \beta_1 X$$

where: Y is the dependent variable X is the independent variable β_0 is the intercept β_1 is the slope

3.2.2 Partial correlation analysis

Partial correlation is a measure of the strength and direction of a linear relationship between two continuous variables whilst controlling for the effect of one or more other continuous variables. The Partial Correlation coefficient $r_{12,3}$ between two variables A1, A2, after removing the influence of the A3, is given as

(Ashok et.,al 2007)

$$r_{12,3} = \frac{r_{12} - r_{13}r_{23}}{\sqrt{1 - r_{13}^2} \sqrt{1 - r_{23}^2}}$$

3.2.3 Composite analysis

Composite analysis is widely used to detect relations between climate phenomenon and potential causal factors. Composite analysis is considered as simple in the sense that only two basic steps are required to form a composite analysis (von Storch and Zwiers, 1999): form sets () of the events and calculate the composite mean values (e.g. Prein et al., 2017). In practice, the second step is often modified by calculating either the composite anomaly (e.g. Ullah et al., 2021) or the composite difference (e.g. Nan et al., 2021). Since the signals of physical mechanisms leading to atmospheric phenomena do exist within the historical data, the accumulation and averaging of successive events can amplify these signals while the stochastic background noise can be substantially reduced (Laken and Calogović, 2013). Composite analysis is thus a simple and powerful approach to detect how a low-amplitude signal finds its expression in other variables. In this study we have used composite patterns by taking positive and negative AOI years. Further, we employed composite patterns of parameters anomalies for four combinations of the AO and El Niño-Southern Oscillation (ENSO).

3.2.4 EOF analysis

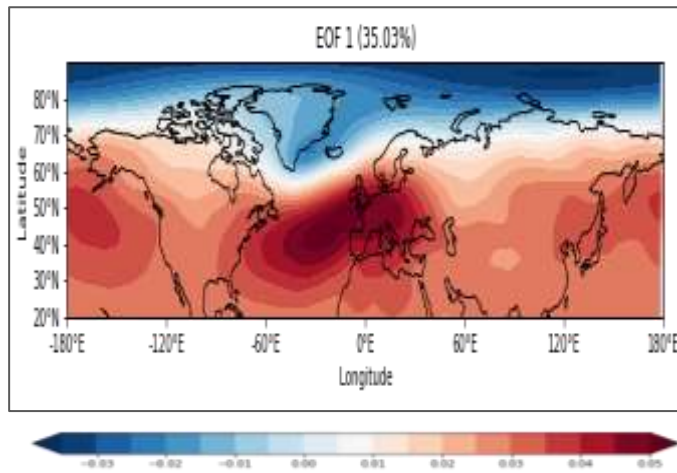
In statistics and signal processing, the method of empirical orthogonal function (EOF) analysis is a decomposition of a signal or data set in terms of orthogonal basis functions which are determined from the data. It is the same as performing a principal components analysis on the data, except that the EOF method finds both time series and spatial patterns. Empirical orthogonal function (EOF) analyses are often used to study possible spatial patterns of climate variability and how they change with time. One of the important results from EOF analysis is the discovery of several oscillations in the climate system, including the Pacific Decadal Oscillation and the Arctic Oscillation. Similarly to Fourier analysis and wavelet analysis, in EOF analysis, one also projects the original climate data on an orthogonal basis. However, this orthogonal basis is derived by computing the eigenvectors of a spatially weighted anomaly covariance matrix, and the corresponding eigenvalues provide a measure of the percent variance explained by each pattern. Therefore, EOFs of a space-time physical process can represent mutually orthogonal space patterns where the data variance is concentrated, with the first pattern being responsible for the largest part of the variance, the second for the largest part of the remaining variance, and so on. Later, in order to overcome some limitations of classical EOF analysis and make the resulting patterns more physically interpretable, rotated EOFs, Hilbert EOFs, and complex EOFs are developed. However, in our study, we have used classical EOF analysis to calculate the Arctic Oscillation Index (AOI).

Chapter 4

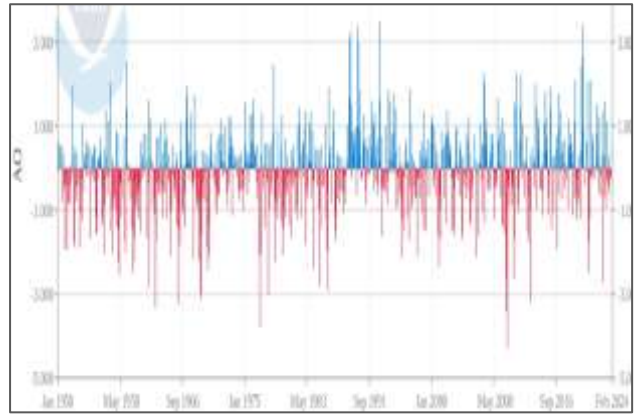
Results and Discussion

To study the relationship of Arctic Oscillation with other climatic modes firstly we have calculated the Arctic Oscillation Index (AOI). It is calculated as the principal component (PC) associated with first leading mode of variability (EOF1; which captures the maximum amount of variance) of monthly mean geopotential height anomalies at 1000-hPa, poleward of 20°N in the Northern hemisphere, normalised with the monthly mean index's standard deviation (Thompson et al., 1998). The EOF is displayed in figure 4.1a and associated PC (normalised) is displayed in figure 4.1c. Meridional seesaw spatial pattern of leading mode of variability (Fig. 4.1a) is characteristic of Arctic oscillation. AOI can also be obtained from the Climate Prediction Centre (CPC) website ([cpc.ncep.gov/products/precip.CWlink./daily_ao_index](https://cpc.ncep.gov/products/precip/CWlink/daily_ao_index)) which is displayed in figure 4.1b from the years 1950-2023. Correlation between AOI calculated by EOF and from the website is 0.99 which confirms the robustness of calculated AOI index. Further, AOI for JJAS season is extracted from the monthly index and used for analysis in the study. To evaluate the relationship of AOI with different parameters during boreal summer season, we have performed correlation and regression analysis as follows.

(a) EOF 1 Pattern



(b) AO Index (Available)



(c) PC1

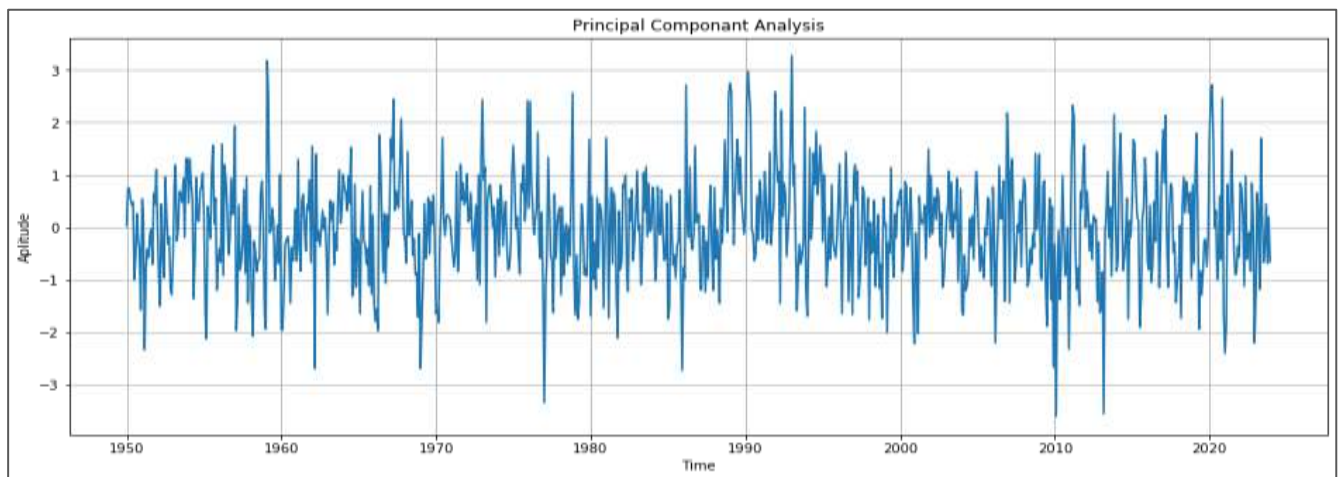


Figure 4.1 displays (a) the leading mode of variability (EOF1) of 1000 hPa monthly geopotential height over north of 20°N for the period 1950-2023 (b) AOI index from CPC website and (c) associated PC with EOF1.

4.1 Relationship of Arctic Oscillation with different climatic modes

Figure 4.2 displays the correlation between AOI and climatic modes like Precipitation, SST, MSLP etc., over the entire northern hemisphere starting from 20°S. Positive correlation in precipitation appears over most of the Indian landmass specifically; positively high correlation is seen over the monsoon trough region and northwestern part of India (Fig 4.2a). Further, the positive correlation is displayed majorly over the northern parts of China, southern parts of Mexico, East coast of USA, and north eastern coast of South America continent while negative correlation appears over central part of USA. It

is concluded that increasing the strength of the AO (positive AO) phase enhances the precipitation over Indian landmass.

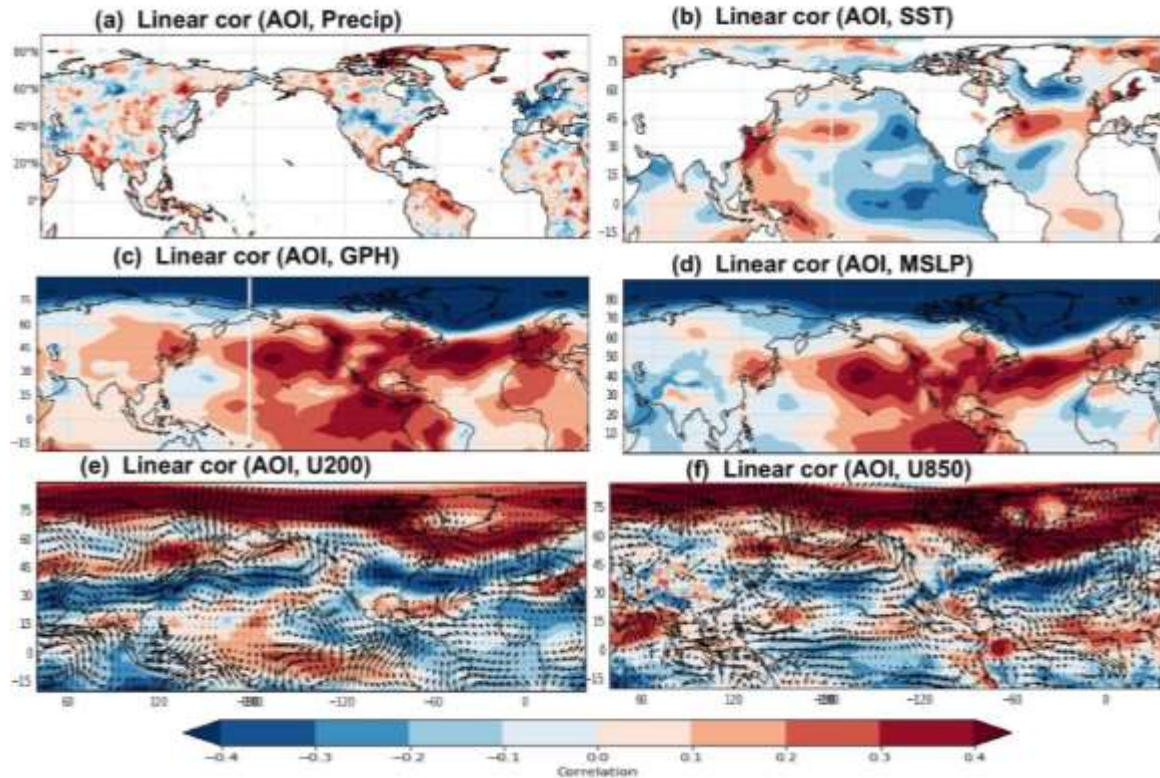


Figure 4.2 shows the linear correlation between Arctic oscillation Index and anomalies of (a) Geopotential height (GPH) (b) Mean Sea Level Pressure (MSLP) (c) Precipitation (d) SST (e) Zonal winds (shaded) and winds (vectors) at 200 hPa and (f) same as (e) but for 850hPa winds.

The correlation with SSTs show that AOI is strongly associated with La-Nina like cooling over the equatorial eastern Pacific with a positive PDO like pattern in the northern Pacific and warming over the western part of Pacific Ocean which is extending from the northern coast of Australia to Western North Pacific (WNP) (Fig. 4.2b). However, very weak SSTs are seen over the entire Indian Ocean. This La-Nina associated with AOI may induce positive rainfall anomalies over India and negative rainfall anomalies over the western coast of South America. Positively high GPH anomalies over midlatitudes associated with AOI and negative over the polar regions depict typical Arctic Oscillation (Fig. 4.2c). At the same time, positive but weak GPH is seen over Indian landmass covering Indian Ocean region and positively high GPH is seen over the tropical eastern and north eastern Pacific which induces the cold and moderate SSTs underneath with suppressing and enhancing the rainfall respectively as observed. Further, high SLP anomalies over eastern

Pacific and low SLP over Indian Ocean extending to Indian landmass is consistent with GPH and SST anomalies (Fig. 4.2d). At the same time, the upper level winds associated with AOI shows that strong northward displacement of Asian jet from its climatological mean position extending from West to East Asia (Fig. 4.2e) under the presence of La Nina, which is consistent with the previous studies. La Nina and positive phase of PDO induces a negative tropospheric temperature (TT) anomalous in tropics to midlatitudes, weakens the meridional TT gradient in the tropics and strengthens it over the midlatitudes thus thereby northward shifting the Asian jet according to thermal wind equation (e.g., Pedlosky, 1979 Sections 2.6, 2.9b, and 6.5). This northward shifted Asian jet and La Nina may induce the low level strong cyclonic circulation over north of India (Fig. 4.2f), strengthening the southwesterly monsoon flow with enhancing rainfall over India. From above correlation analysis it is confirmed that AO has strong association with the SSTs over the Pacific and Asian jet along with ISM rainfall. As discussed earlier, AO exhibits positive and negative phases, to separately analyse the influence of these phases on ISM rainfall and large circulation, we have conducted a composite analysis for the study period 1950 to 2023 which is discussed in next section.

4.2 Influence of Positive and Negative phase of AO on ISM rainfall

For that, we separated the positive and negative AO years based on AOI Index values. Then we assign Niño 3.4 Index values for the same years so we got nearly 15 Positive AOI years and 19 Negative AOI years. Later we took a Composite of separate positive and negative years to examine the variability similarly composite was taken for all Used Meteorological variables such as Geopotential Height, Mean Sea Level Pressure, Land Temperature, Precipitation, Sea Surface Temperature.

Positive Years	Niño 3.4	Negative Years	Niño 3.4
1953	0.5369	1951	0.87056
1955	-1.9364	1957	1.033
1961	-0.402	1958	0.10983
1967	-0.455	1960	-0.1708
1970	-1.224	1963	0.972
1973	-1.875	1965	1.533
1975	-2.070	1968	0.3579
1983	-0.048	1972	1.619
1988	-2.347	1977	0.6102
1989	-0.842	1987	2.2740
2006	0.5491	1993	0.565
2016	-0.45562	1998	-1.12
2018	0.3010	2008	-0.32
2021	-0.463	2009	1.0088
		2011	-0.77
		2014	0.455
		2015	2.733
		2019	0.5573
		2022	-1.362

La Nina(< -0.5)	Neutral (+0.5,-0.5)	El Niño (> +0.5)	Neutral(+0.5,-0.5)
1955: -1.9364 1970: -1.224 1973: -1.875 1975:-2.070 1983: -0.048 1988: -2.347 1989: -0.842	1953: 0.5369 1961: -0.402 1967: -0.455 1994: 0.54 2006: 0.455 2016: 0.455 2018: 0.3010 2021: -0.463	1951: 0.87056 1957: 1.033 1963: 0.972 1965: 1.533 1972: 1.619 1977: 0.6102 1987: 2.2740 1993: 0.565 2009: 1.0088 2015: 2.733 2019: 0.5573 23	1958: 0.10983 1960: -0.1708 1968: 0.3579 2008: -0.32

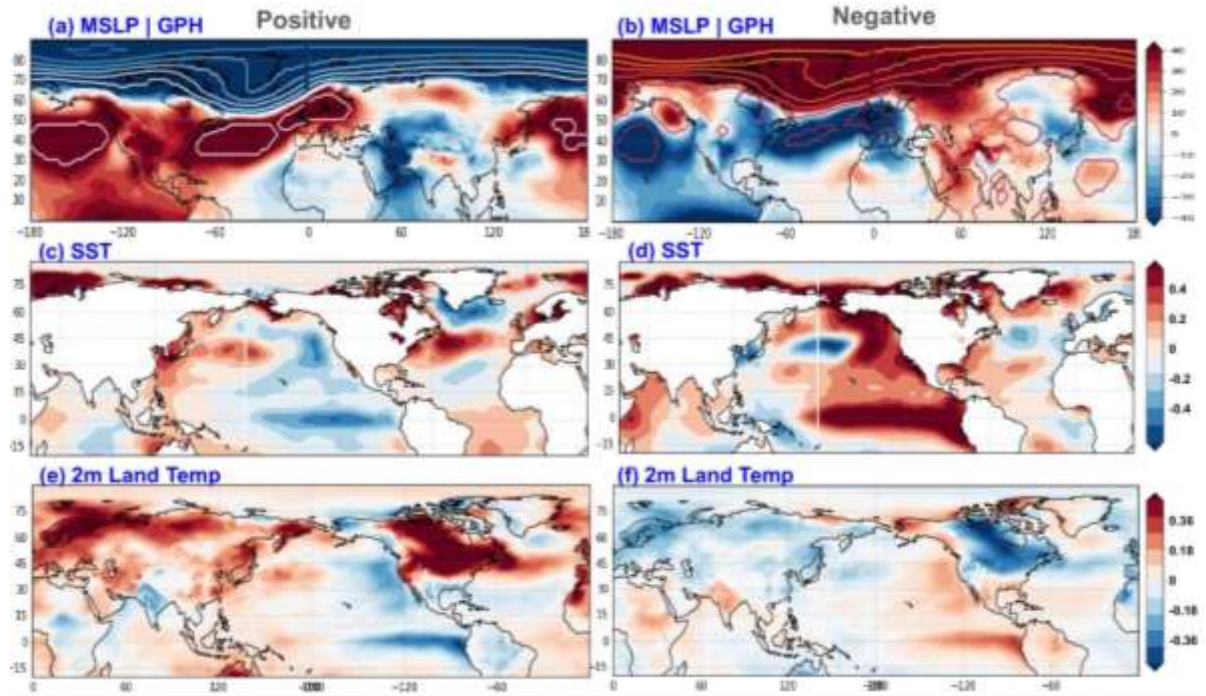


Figure 4.3 displays the composite patterns of a) MSLP (shaded, hPa) and GPH (contours; gpm), c) SST (°C) and e) 2m air temperature (°C) anomalies for positive AOI years. Figs (b, d, f) same as with (a, c, e) respectively but for negative AOI years.

Figure 4.3 show the composite analysis of various weather parameters such as sea level pressure, geopotential height, SST, 2-meter land temperature, and wind circulation for 200 hPa and 850 hPa pressure levels for the positive and negative years of Arctic Oscillation. Figures 4.3a and b exhibit the composite analysis of sea level pressure with geopotential height composite and consistent with sea level pressure. Figures 4c and d interpret the positive AO year show strong La Nina conditions and negative AO years display strong El Niño years suggesting the strong association between AO and ENSO. Low-level wind for the positive years showing strong low level jet over the Arabian Sea and a cyclonic circulation over Indian subcontinent, which favours the positive precipitation anomalies (Fig. 4.4a and c). In case of negative AO years shows an low level anticyclone over India and weak low level jet over the Arabian Sea and as a result reduced ISM rainfall is seen (Fig. 4.4d and f). All these results are in line with regression analysis as discussed earlier.

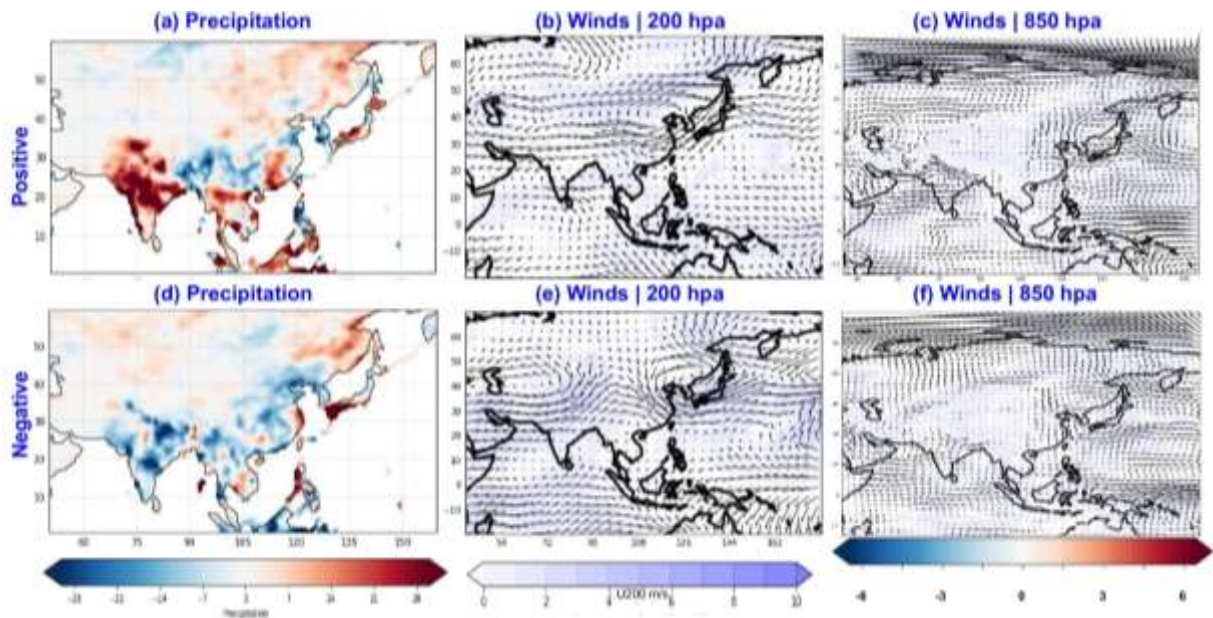


Figure 4.4 shows the composite patterns of a) Precipitation (mm), b) Zonal winds (shaded; m/s), winds (vectors; m/s) at 200 hPa c) Zonal winds (shaded; m/s), winds (vectors; m/s) at 850 hPa for positive AOI years. Figs (d, e, f) same as with (a, b, c) respectively but for negative AOI years.

It is noticed that positive (negative) AO is accompanied by La Niña (El Niño) events, combined impact of AO and ENSO shows strong impact on ISM rainfall. However, it is essential to isolate the impact of ENSO and study the AO only influence on ISM rainfall. Based on the JJAS Niño 3.4 index and AO index, we have classified AO years into four categories 1. Positive AO and La Nina (AO^+La) years, 2. Negative AO and El Niño (AO^-El), 3. Positive AO and ENSO neutral conditions years (AO^+) and Negative AO and ENSO neutral conditions years (AO^-) and performed composite analysis.

4.3 Isolating the impact of different phases of AO and ENSO on ISM rainfall

Figure 4.5 shows SST anomalies during AO^+La , AO^-El , AO^+ and AO^- composite years. During AO^+La (AO^-El) years strong negative (positive) SST anomalies are apparent over the central and equatorial Pacific region, north Pacific, north Atlantic and parts of Arctic Ocean. In case of AO^+ and AO^- years, warm SST anomalies are seen only over the northern parts of Pacific and Atlantic Ocean.

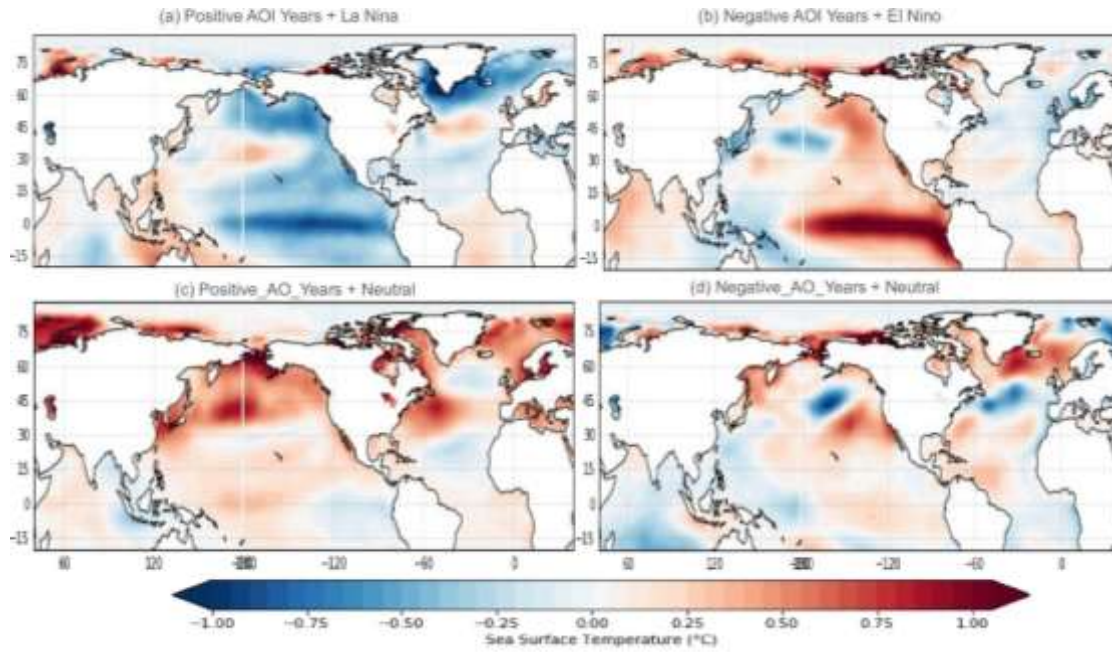


Figure 4.5 displays the composite patterns of SST ($^{\circ}C$) anomalies for four combinations of the AO and El Niño-Southern Oscillation (ENSO) which are the (a) Positive AO and La Niña (AO^+La) b) Negative AO and El Niño (AO^-El) c) Positive AO and neutral ENSO (AO^+) and d) Negative AO and neutral ENSO years (AO^-).

SLP anomalies during AO^+La (AO^-El) years show strong negative (positive) anomalies over the northern polar region and positive (negative) in the mid-latitude region (Fig. 4.6a and b). At the same time negative (positive) SLP anomalies also observed over ISM region in AO^+La (AO^-El) years. These strong SLP anomalies over India are due to both ENSO and AO forcing. For the AO^+ SLP anomalies are weak negative over India and strong negative in and AO^- (Fig. 4.6c and d). GPH anomalies are consistent with SLP anomalies in all four composites.

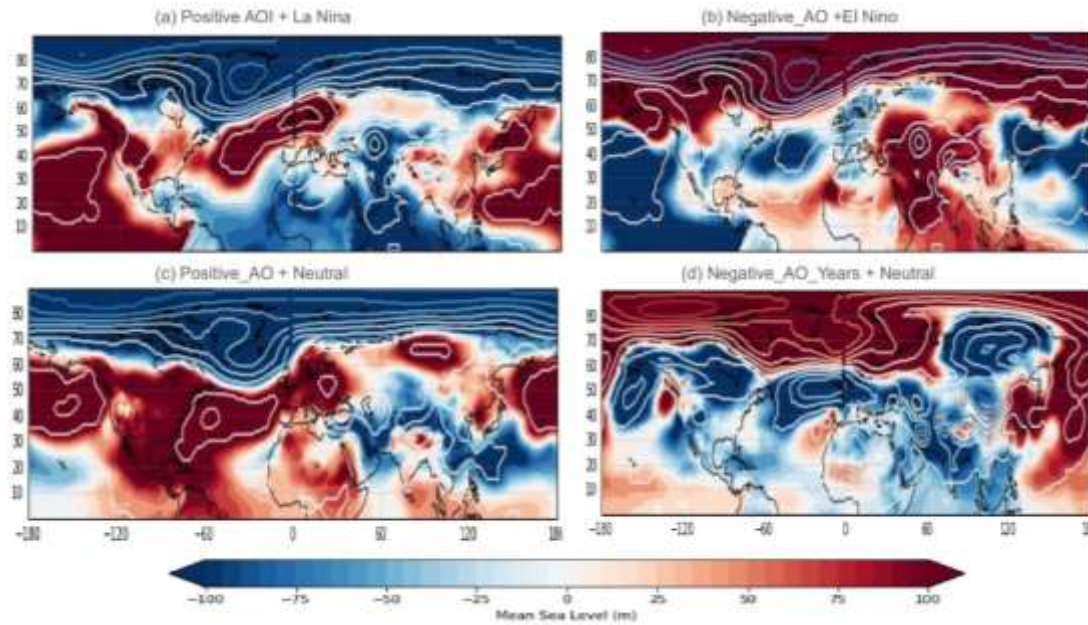


Figure 4.6 illustrates the Composite patterns of MSLP (shaded, m) and GPH (contours; gpm) anomalies for four combinations of the AO and ENSO which are the (a) Positive AO and La Niña b) Negative AO and El Niño c) Positive AO and neutral ENSO d) Negative AO and neutral ENSO years.

During AO⁺La years precipitation anomalies over most part of India is positive, which is contributed by La Nina like conditions in the Pacific in addition to AO⁺ (Fig. 4.7a). On the other hand, AO⁻El years shows negative precipitation anomalies over India as a result of both AO and El Niño forcing (Fig. 4.7b). Similarly, strong positive precipitation anomalies are seen over India during AO⁺ years, indicting the positive AO alone could induce the strong influence on ISM rainfall. During AO⁻ years, a weak negative precipitation anomaly over Indian region is apparent. The positive precipitation anomalies are in phase with negative SAT anomalies over India in both AO⁺La and AO⁺ years (Fig. 4.8a and c). Similarly, the negative precipitation anomalies are in line with warm SAT over India during AO⁻La and AO⁻ years (Fig. 4.8b and d). It is essential to understand the mechanisms that links the AO and ISM rainfall with and without ENSO influence.

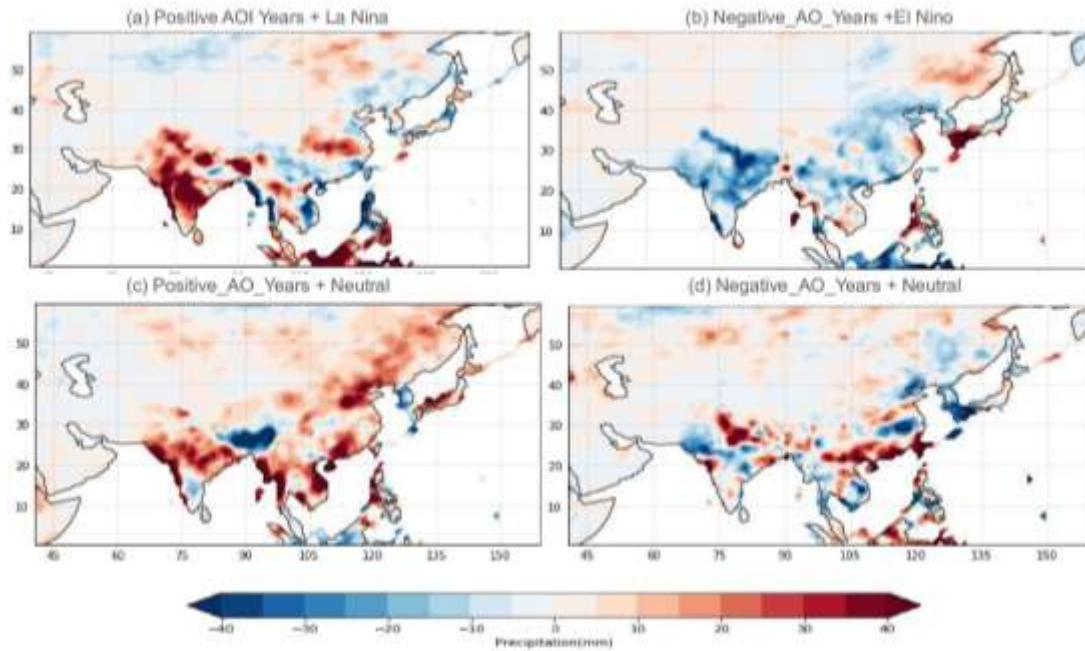


Figure 4.7 displays the composite patterns of Precipitation (mm) anomalies for four combinations of the AO and ENSO which are the (a) Positive AO and La Niña b) Negative AO and El Niño c) Positive AO and neutral ENSO d) Negative AO and neutral ENSO years.

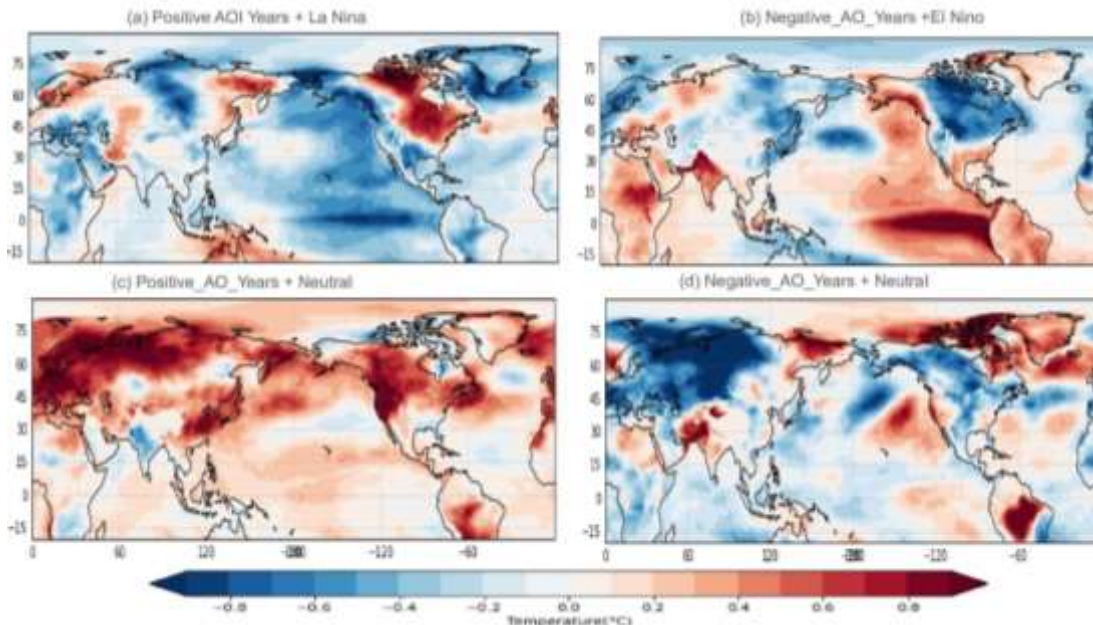


Figure 4.8 shows the composite patterns of atmospheric temperature at 2 meter (°C) anomalies for four combinations of the AO and ENSO which are the (a) Positive AO and La Niña b) Negative AO and El Niño c) Positive AO and neutral ENSO d) Negative AO and neutral ENSO years.

Previous studies revealed that AO directly alters the strength and latitudinal position of the polar jet and subtropical westerly jet (He et al., 2017; Mao et al., 2011a; Mao et al., 2011b). During positive AO years a strong low-pressure system near poles and a strong high-pressure system (Song and Wu 2018) in the mid-latitudes led to a stronger and conventional polar jet and northward shift of the westerly jet (Cheung et al., 2016). As a result of this, cold Arctic air is restricted to polar region. In general, northward shift of the westerly jet over the Asian region called Asian Jet results in strong ISM rainfall. As the Asian jet migrates to poleward side favours the ISM circulation to strengthen at low levels and enhances the monsoon rainfall. As a result of this mechanism, a strong cyclonic circulation over India with strong south-westerly anomalies are seen in AO⁺ years (Fig. 4.9c and 4.10c). In contrast in AO⁻ years, southward displacement of Asian jet is seen (Fig. 4.9d) and this southward displaced jet reinforce the low level circulation over ISM region and as a result weak anticyclonic circulation causes negative precipitation over India (Fig. 4.10d).

In AO⁺La years, in addition to AO influence La Nina related forcing through the Walker circulation with ascending branch over ISM region could cause strong positive precipitation anomalies over India (Fig 4.7a, Fig 4.9a and Fig 4.10a). Southward displacement of Asian jet is much stronger during El Niño years when co-occurred with AO as indicated in Fig 4.9b. In addition to this southward displacement of Asian jet, strong El Niño related subsidence causes negative precipitation anomalies over India (Fig. 4.7b), which is well reflected in the weak low level circulation.

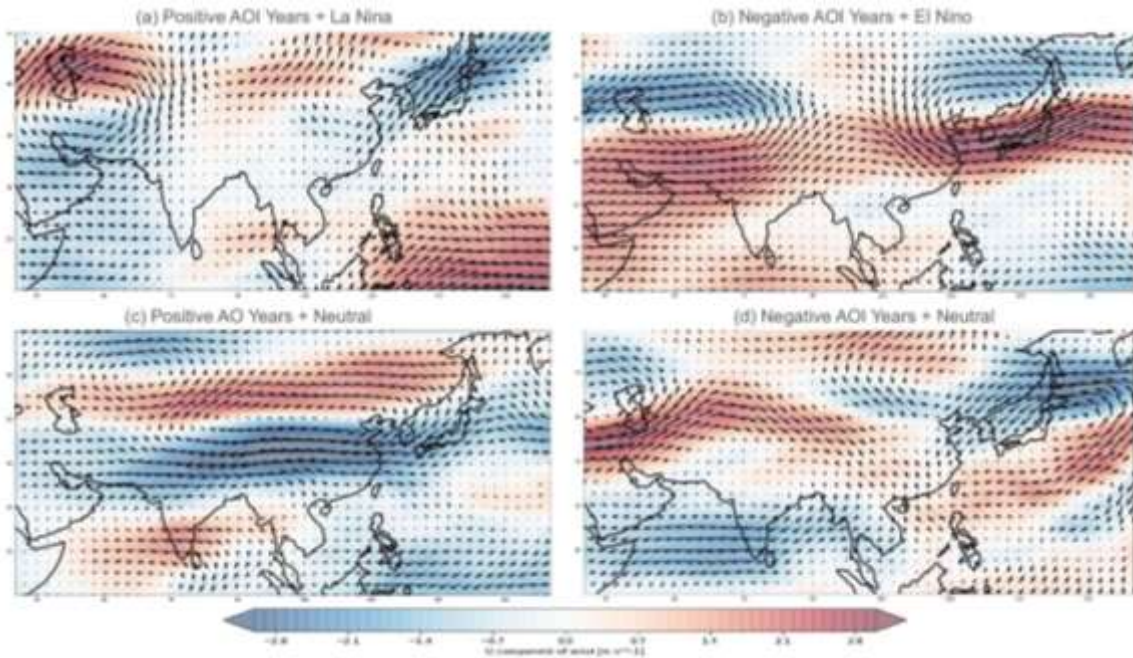


Figure 4.9 illustrates the composite patterns of zonal wind (shaded; m/s), winds (vectors; m/s) at 200 hPa anomalies over Asian region for four combinations of the AO and ENSO which are the (a) Positive AO and La Niña b) Negative AO and El Niño c) Positive AO and neutral ENSO d) Negative AO and neutral ENSO years.

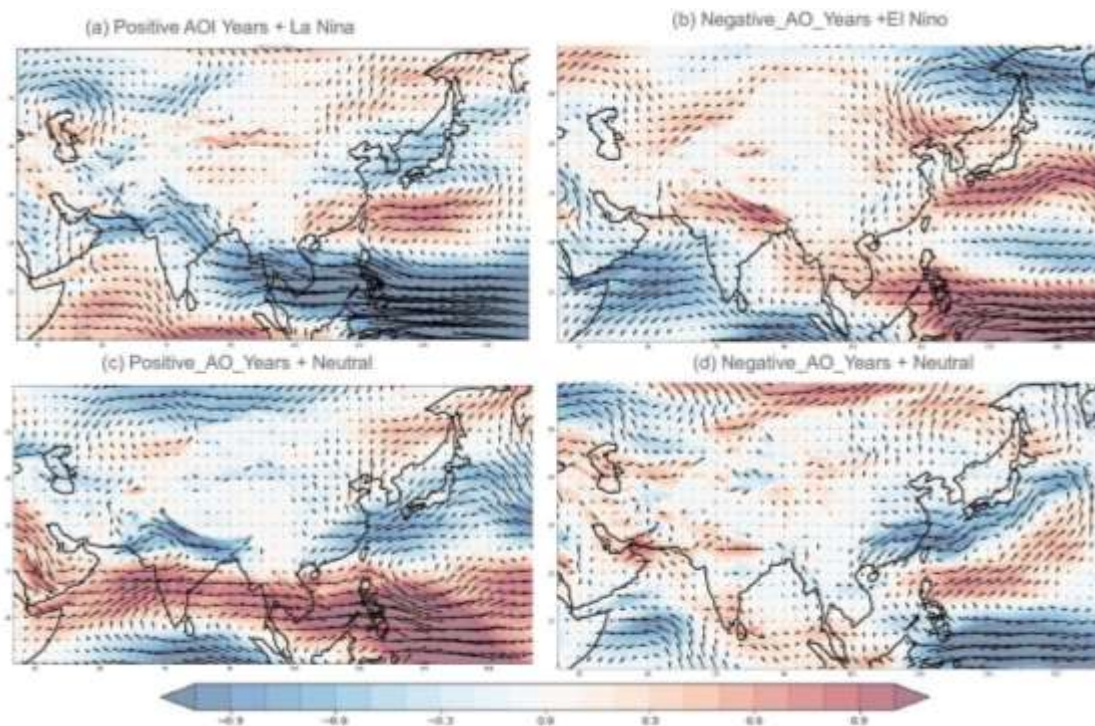


Figure 4.10 shows the composite patterns of zonal wind (shaded; m/s), winds (vectors; m/s) at 850 hPa anomalies for four combinations of the AO and ENSO which are the (a) Positive AO and La Niña b) Negative AO and El Niño c) Positive AO and neutral ENSO d) Negative AO and neutral ENSO years.

Over all, this analysis suggests that AO co-occurred with ENSO strongly influence the ISM rainfall, in addition to this summer positive (negative) AO without ENSO causes (negative) positive rainfall anomalies over India through meridional displacement of Asian jet and altering the low level ISM circulation.

Chapter 5

Summary and Conclusion

Indian Summer monsoon rainfall is known to be altered by external drivers or climate modes such as ENSO, IOD and Atlantic Niño etc from tropics and NAO and Eurasian snow cover (GIVE references) on the interannual time scale. A few studies indicated that the Arctic Oscillation (AO) could influence the ISM rainfall. However, role of summer (AO) impact on ISM rainfall is not well addressed. In this study impact of boreal summer AO on ISM rainfall variability is examined. Combined impact of AO and ENSO and individual impact of AO positive and negative phase on ISM rainfall is the main focus of the study.

Detailed analysis suggest that the positive AO without neutral ENSO conditions displace the Asian jet northward allow the ISM circulation at low level strengthens and enhance the rainfall over India. In contrast, negative AO with neutral ENSO causes southward displacement of Asian jet and it weakens the monsoon circulation and caused for negative rainfall anomalies over India.

In case of positive AO co-occurred with La Nina events, changes in Walker circulation with ascending limb over ISM region in addition to northward displacement of Asian jet causes enhanced rainfall over India. Similarly, when negative AO co-occurred with El Niño, descending branch of Walker circulation in addition to southward displacement of Asian jet causes reduced rainfall over India. These results are useful to better understand the ISM interannual variability and predictability.

Chapter 6

Future scope

- ✓ To understand the relation between the Arctic Oscillation and ISM rainfall variability in CMIP6 models.
- ✓ To investigate the decadal variability of AO and its influence on ISM rainfall and associated mechanisms.

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